# Logical operators binary truth table definition

| Input | | Output | | | | |
|---|---|---|---|---|---|---|
| | | Conjunction | Exclusive or | Disjunction | Conditional | Biconditional |
| $p$ | $q$ | $p \wedge q$ | $p \oplus q$ | $p \vee q$ | $p \to q$ | $p \leftrightarrow q$ |
| $T$ | $T$ | $T$ | $F$ | $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ | $T$ | $T$ | $F$ | $F$ |
| $F$ | $T$ | $F$ | $T$ | $T$ | $T$ | $F$ |
| $F$ | $F$ | $F$ | $F$ | $F$ | $T$ | $T$ |
| | | "$p$ and $q$" | "$p$ xor $q$" | "$p$ or $q$" | "if $p$ then $q$" | "$p$ if and only if $q$" |

# Logical operators truth tables

Truth tables: Input-output tables where we use $T$ for 1 and $F$ for 0.

| Input | | Output | | |
|---|---|---|---|---|
| | | **Conjunction** | **Exclusive or** | **Disjunction** |
| $p$ | $q$ | $p \wedge q$ | $p \oplus q$ | $p \vee q$ |
| $T$ | $T$ | $T$ | $F$ | $T$ |
| $T$ | $F$ | $F$ | $T$ | $T$ |
| $F$ | $T$ | $F$ | $T$ | $T$ |
| $F$ | $F$ | $F$ | $F$ | $F$ |
| | |  |  |  |







| Input | Output |
|---|---|
| | **Negation** |
| $p$ | $\neg p$ |
| $T$ | $F$ |
| $F$ | $T$ |
| |  |



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